

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Keerthikaa N ¹⁹²⁵²¹⁰⁶⁴ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Keerthikaa

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

2) Subnetting Analysis for Two systems

Aim:

To determine the subnet range, network address, broadcast address, valid host IP range, and the number of usable host addresses for two systems using the given subnet mask.

Given data:

* System 1 IP Address : 192.168.1.130

* System 2 IP Address : 192.168.1.190

* Subnet mask : 255.255.255.192

* CIDR Notation : /26

Theory:

Subnetting is the process of dividing a large network into smaller logical networks called subnets. It helps improve network performance, enhances security and makes efficient use of IP address. A subnet mask identifies the network portion and the host portion of an IP address.

The subnet mask 255.255.255.192 is equivalent to /26 which means that 26 bits are used for the network portion and 6 bits are used for the host portion.

The number of usable host addresses is calculated using the formula:

$$\text{Usable Hosts} = 2^{(\text{Host Bits})} - 2$$

$$= 2^6 - 2$$

$$= 64 - 2$$

$$= 62 \text{ usable host addresses}$$

The block size is calculated as :

$$\text{Block Size} = 256 - 192 = 64$$

Therefore, the subnet ranges are :

$$* 192.168.1.0 - 192.168.1.63$$

$$* 192.168.1.64 - 192.168.1.127$$

$$* 192.168.1.128 - 192.168.1.191$$

$$* 192.168.1.192 - 192.168.1.255$$

System 1 Analysis:

IP Address : 192.168.1.130

The IP Address 192.168.1.130 lies in the subnet
192.168.1.128 - 192.168.1.191

$$* \text{Network Address : } 192.168.1.128$$

$$* \text{First Valid host address : } 192.168.1.129$$

$$* \text{Last Valid host address : } 192.168.1.190$$

$$* \text{Broadcast Address : } 192.168.1.191$$

$$* \text{Usable host Address : } 62$$

Since 192.168.1.130 lies between 192.168.1.129 and 192.168.1.190, the IP Address belongs to this subnet and is a valid host address.

System 2 Analysis:

IP Address : 192.168.1.190

The IP address also falls in the subnet 192.168.1.128 -
192.168.1.191

$$* \text{Network Address : } 192.168.1.128$$

$$* \text{First Valid host address : } 192.168.1.129$$

$$* \text{Last valid host address : } 192.168.1.190$$

* Broadcast Address : 192.168.1.191

* Usable host addresses : 62

The IP address 192.168.1.190 is the last usable host address in the subnet and is therefore valid.

Conclusion:

Using the subnet mask 255.255.255.192 (/26), both systems are found to be in the same subnet. Hence both IP addresses are valid and each subnet can accommodate 62 usable host devices.

3) Subnet Allocation for Different Departments

Aim:

To allocate subnet addresses for different departments using the given IP block and subnet masks. Also determine the network address, valid host IP range, broadcast address and number of usable host for each department.

Given data:

* Network Address : 172.16.0.0/16

Departments:

* HR → /25

* Sales → /26

* IT → /27

* Admin → /28

Theory:

Subnetting divides a large network into smaller subnetworks for efficient IP address utilization and improved network performance. The subnet mask determines the network and host bits. Larger departments receive larger subnets, while smaller departments receive smaller subnets to avoid wasting IP addresses.

HR department (/25)

- * Network address : 172.16.0.0
- * Subnet mask : 255.255.255.128
- * Host range : 172.16.0.1 - 172.16.0.126
- * Broadcast address : 172.16.0.127
- * Usable Hosts : 126

Sales department (/26)

- * Network address : 172.16.0.128
- * Subnet mask : 255.255.255.192
- * Host range : 172.16.0.129 - 172.16.0.190
- * Broadcast Address : 172.16.0.191
- * Usable hosts : 62

IT department (/27)

- * Network address : 172.16.0.192
- * Subnet mask : 255.255.255.224
- * Host range : 172.16.0.193 - 172.16.0.222

* Broadcast address : 172.16.0.223

* Usable hosts : 30

Admin department (128)

* Network address : 172.16.0.224

* Subnet mask : 255.255.255.240

* Host range : 172.16.0.225 - 172.16.0.238

* Broadcast address : 172.16.0.239

* Usable hosts : 14

Conclusion :

The subnets were successfully allocated for the HR, sales, IT and admin departments using the given subnet masks. This subnetting scheme improves network organisation, reduces unnecessary IP wastage and provides efficient communication between devices.